

Some (potentially) helpful formulas:

Thermodynamics:

$$dE = dQ + dW$$

$$C_X = \left. \frac{dQ}{dT} \right|_X$$

$$dE = TdS + \sum_i J_i dx_i + \mu dN$$

$$F = E - TS$$

$$G = E - TS - Jx$$

$$\Phi = E - TS - \mu N$$

Probability/Mathematical Formulas:

$$\tilde{p}(k) = \langle e^{-ikx} \rangle = \sum_{n=0}^{\infty} \frac{(-ik)^n}{n!} \langle x^n \rangle$$

$$\ln \tilde{p}(k) = \sum_{n=1}^{\infty} \frac{(-ik)^n}{n!} \langle x^n \rangle_c$$

$$\int_0^{\infty} dx x^n e^{-\alpha x} = \frac{n!}{\alpha^{n+1}}$$

$$\int_{-\infty}^{\infty} dx e^{-ikx - \frac{x^2}{2\sigma^2}} = \sqrt{2\pi\sigma^2} e^{-\frac{\sigma^2 k^2}{2}}$$

$$\int_{-\infty}^{\infty} x^2 e^{-ax^2} = \frac{1}{2} \sqrt{\frac{\pi}{a^3}}$$

$$\lim_{x \rightarrow \infty} N! = N \ln N - N$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n!}$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

$$\sinh x = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$$

$$\cosh x = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

Kinetic Theory of Gases

Student #4

$$\frac{d\vec{q}_i}{dt} = \frac{\partial H}{\partial \vec{p}_i}$$

$$\frac{d\vec{p}_i}{dt} = -\frac{\partial H}{\partial \vec{q}_i}$$

$$\{A, B\} = \sum_{\alpha=1}^{3N} \left(\frac{\partial A}{\partial q_{\alpha}} \cdot \frac{\partial B}{\partial p_{\alpha}} - \frac{\partial A}{\partial p_{\alpha}} \cdot \frac{\partial B}{\partial q_{\alpha}} \right)$$

$$\frac{\partial \rho}{\partial t} = \{H, \rho\}$$

Classical Statistical Mechanics

$$p_{(T,x,N)}(\mu_s) = \frac{e^{-\beta H(\mu_s)}}{Z(T, x, N)}$$

$$Z(T, x, N) = \sum_{\mu_s} e^{-\beta H(\mu_s)}$$

$$p_{(T,J,N)}(\mu_s) = \frac{e^{\beta \cdot J \cdot x - \beta H(\mu_s)}}{Z(T, J, N)}$$

$$Z(T, J, N) = \sum_{\mu_s} e^{-\beta(H(\mu_s) - Jx)}$$

$$p_{(T,\mu,x)}(\mu_s) = \frac{e^{\beta \cdot J \cdot x - \beta H(\mu_s)}}{Z(T, J, N)}$$

$$Q(T, \mu, x) = \sum_{\mu_s} e^{-\beta(H(\mu_s) - \mu N)}$$

$$S = k \ln \Omega$$

$$F = -k_B T \ln Z(T, x, N)$$

$$G = -k_B T \ln Z(T, J, N)$$

$$\Phi = -k_B T \ln Q(T, \mu, x)$$

$$\langle E \rangle = -\frac{\partial \ln Z}{\partial \beta}$$

$$\langle x \rangle = k_B T \frac{\partial \ln Z}{\partial J}$$

$$\langle N \rangle = \frac{\partial \ln Q}{\partial (\beta \mu)}$$

$$\lambda(T) = \frac{h}{\sqrt{2\pi m k_B T}}$$

Interacting Particles

$$\ln Z = \ln Z_0 + \frac{N(N-1)}{2V} \int d^3\vec{q} [\exp(-\beta V(\vec{q})) - 1] + O\left(\frac{N^3}{V^3}\right)$$

$$\frac{P}{k_B T} = \frac{N}{V} \left[1 + B_2(T) \frac{N}{V} + B_3(T) \frac{N^2}{V^2} + \dots \right]$$

$$f(\vec{q}) = \exp(-\beta V(\vec{q})) - 1$$

$$B_l(T) = -\frac{l-1}{l!} \bar{d}_l$$

$$(P - an^2)(V - Nb) = Nk_B T$$

$$Z(T,N,V) \approx \frac{(V - N\Omega/2)^2}{N! \lambda^{3N}} \exp \frac{\beta \mu N^2}{2V}$$

Quantum Statistical Mechanics

$$\eta = \begin{cases} +1 & \text{for bosons} \\ -1 & \text{for fermions} \end{cases}$$

$$\ln Q_\eta = -\eta \sum_{\vec{k}} \ln \left[1 - \eta \exp \left(\beta \mu - \beta \epsilon(\vec{k}) \right) \right]^{-1}$$

$$z = e^{\beta \mu}$$

$$N_\eta = \sum_{\vec{k}} \frac{1}{z^{-1} e^{\beta \epsilon(\vec{k})} - \eta}$$

$$E_\eta = \sum_{\vec{k}} \frac{\epsilon(\vec{k})}{z^{-1} e^{\beta \epsilon(\vec{k})} - \eta}$$

$$\beta P_\eta = \frac{g}{\lambda^3} f_{5/2}^\eta(z)$$

$$n_\eta = \frac{g}{\lambda^3} f_{3/2}^\eta(z)$$

$$\epsilon_\eta = \frac{3}{2} P_\eta$$

$$f_m^\eta(z) = \frac{1}{(m-1)!} \int_0^\infty \frac{dx}{x^{m-1}} z^{-1} e^x - \eta = \sum_{\alpha=1}^\infty \eta^{\alpha+1} \frac{z^\alpha}{\alpha^m}$$

$$\frac{d}{dz} f_m^\eta(z) = \frac{1}{z} f_{m-1}^\eta(z)$$

Problem 1 (35 points)

Consider a polymer formed by connecting N disc-shaped molecules into a one-dimensional chain. Each molecule can align along either its long axis (of length $2a$) or short axis (of length a). Furthermore, when laying on its long axis there are two configuration it may assume, whereas when lying along its short axis there is only one possible configuration. The energy of the monomer aligned along its shorter axis is higher by ϵ , that is, the total energy of the system is $H = \epsilon U$ where U is the number of monomers standing up.

a (10 points) Calculate the partition function $Z(T, N)$ of the polymer.

$$2 \quad Z = \frac{1}{N! h^N} \int \prod_{i=1}^N dp_i dq_i e^{-\beta H} = \frac{1}{N! h^N} \int \prod_{i=1}^N dp_i dq_i e^{-\beta \epsilon U}$$

$$\Omega = \frac{N!}{N_r (N - N_r)!}$$

For shorter axis (of length a), the energy is ϵ , $E = \epsilon \sum n_i$

For longer axis (of length $2a$), it has no energy

b (15 points) Calculate the average length $\langle L(T, N) \rangle$ and the variance of the length $\langle L(T, N)^2 \rangle_c$ of the polymer.

$$0 \quad \lambda(T) = \frac{h}{\sqrt{2\pi m k_B T}}$$

$$\frac{h}{\sqrt{2\pi m k_B T}} \geq L$$

$$L \leq \frac{h}{\sqrt{2\pi m k_B T}}$$

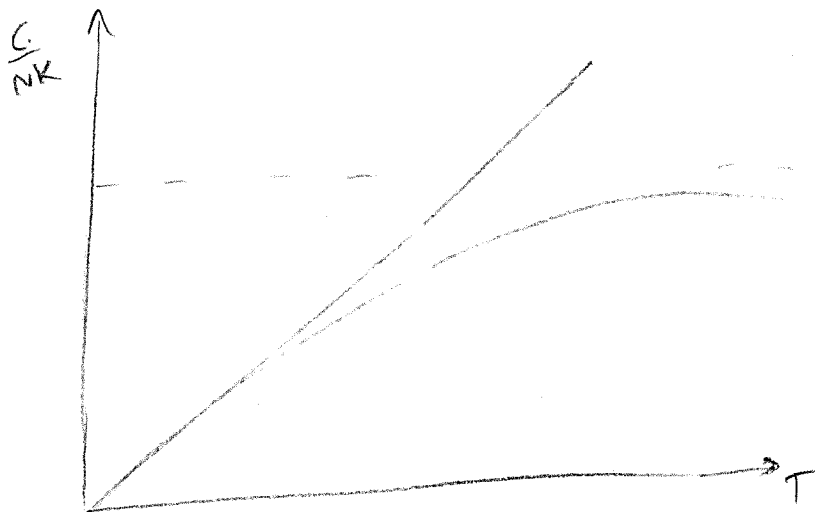
$$\text{and } L^2 \leq \frac{h^2}{2\pi m k_B T}$$

c (10 points) Determine the energy per monomer and heat capacity per monomer. Sketch the heat capacity as a function of temperature.

Energy, $\langle E \rangle = -\frac{\partial}{\partial \beta} \ln Z$

Heat capacity, $C = \frac{\partial \langle E \rangle}{\partial T}$

These are solved for the right partition functions



Problem 2 (35 points)

The pressure P of a gas is related to its density $n = N/V$ and temperature T by the truncated expansion:

$$P = k_B T n - \frac{b}{2} n^2 + \frac{c}{6} n^3$$

where b and c are assumed to be positive temperature-independent constants.

- a (15 points) Locate the critical temperature T_c below which the equation must be invalid, and the corresponding density n_c and pressure P_c of the critical point. Find the ratio $k_B T_c n_c / P_c$.

At critical temperature, $\frac{dP}{dn} = 0$ and $\frac{d^2P}{dn^2} = 0$

$$\frac{dP}{dn} = k_B T_c - \frac{2b}{2} n_c + \frac{3c}{6} n_c^2 = 0 \quad \text{and} \quad \frac{d^2P}{dn^2} = -b + \frac{2c}{2} n_c = 0$$

$$\frac{dP}{dn} = k_B T_c - b n_c + \frac{c}{2} n_c^2 = 0$$

$$-b + c n_c = 0$$

$$c n_c = b$$

$$n_c = \frac{b}{c}$$

$$k_B T_c - b \cdot \frac{b}{c} + \frac{c}{2} \frac{b^2}{c^2} = 0$$

$$k_B T_c - \frac{b^2}{c} + \frac{b^2}{2c} = 0$$

$$k_B T_c - \frac{2b^2 - b^2}{2c} = 0$$

$$k_B T_c - \frac{b^2}{2c} = 0$$

$$k_B T_c = \frac{b^2}{2c}$$

$$T_c = \frac{b^2}{2k_B c}$$

$$P_c = k_B T_c n_c - \frac{b}{2} n_c^2 + \frac{c}{6} n_c^3$$

$$P_c = \frac{k_B b^2}{2c} \frac{b}{c} - \frac{b}{2} \frac{b^2}{c^2} + \frac{c}{6} \frac{b^3}{c^3}$$

$$P_c = \frac{b^3}{2c^2} - \frac{b^3}{2c^2} + \frac{c}{6} \frac{b^3}{c^3}$$

$$P_c = \frac{c}{6} \frac{b^3}{c^3} = \frac{b^3}{6c^2}$$

$$\frac{k_B T_c n_c}{P_c} = \frac{\frac{k_B b^2}{2c} \frac{b}{c}}{\frac{b^3}{6c^2}} = 3$$

$$\frac{k_B T_c n_c}{P_c} = 3$$

b (10 points) Calculate the isothermal compressibility $\kappa_T = -\frac{1}{V} \frac{\partial V}{\partial P} \Big|_T$ and sketch its behavior as a function of T for $n = n_c$.

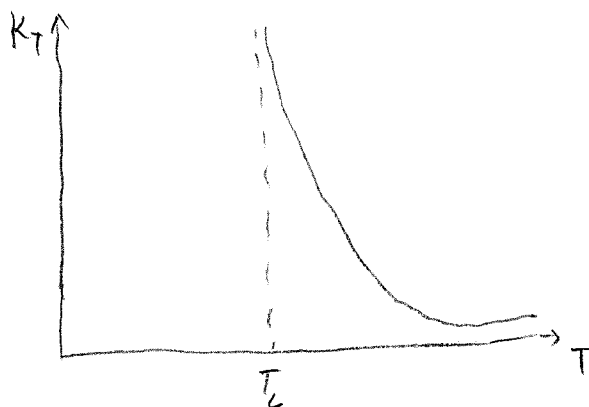
$$\kappa_T = -\frac{1}{V} \frac{\partial V}{\partial P} \Big|_T = -\frac{1}{V} \frac{\partial \left(\frac{N}{n} \right)}{\partial P} \Big|_T = -\frac{1}{n} \frac{\partial \left(\frac{N}{V} \right)}{\partial P} \Big|_T = -\frac{1}{n} \frac{\partial n}{\partial P} \Big|_T = -n^{-1} \left(\frac{\partial P}{\partial n} \right)^{-1} \Big|_T = - \left[n \left(\frac{\partial P}{\partial n} \right) \right]^{-1} \Big|_T$$

$$\frac{\partial P}{\partial n} = k_B T - b n + \frac{c}{2} n^2$$

$$\kappa_T = - \left[n \left(k_B T - b n + \frac{c}{2} n^2 \right) \right]^{-1} \Big|_T$$

$$\text{If } n = n_c, \kappa_T \propto (T - T_c)^{-1}$$

κ_T diverges as $T \rightarrow T_c$



c (10 points) On the critical isotherm give an expression for $(P - P_c)$ as a function of $(n - n_c)$

For critical isotherm,

$$P - P_c = k_B T_c n - \frac{b}{2} n^2 + \frac{c}{6} n^3 - \frac{b^3}{6c^2}$$

$$P - P_c = \frac{k_B b^2}{2k_B c} n - \frac{b}{2} n^2 + \frac{c}{6} n^3 - \frac{c b^3}{6 c^3}$$

$$P - P_c = \frac{b^2}{2c} n - \frac{b}{2} n^2 + \frac{c}{6} n^3 - \frac{c b^3}{6 c^3}$$

$$P - P_c = \frac{c}{6} \left(\frac{3b^2}{c^2} n - \frac{3b}{c} n^2 + n^3 - \frac{b^3}{c^3} \right)$$

$$P - P_c = \frac{c}{6} \left[n^3 - 3 \left(\frac{b}{c} \right) n^2 + 3 \left(\frac{b}{c} \right)^2 n - \left(\frac{b}{c} \right)^3 \right]$$

$$P - P_c = \frac{c}{6} \left[n^3 - 3 n_c n^2 + 3 n_c^2 n - n_c^3 \right]$$

$$P - P_c = \frac{c}{6} (n - n_c)^3 \quad \text{from binomial expansion}$$

Problem 3 (30 points + 10 bonus)

Molecular hydrogen exists in two states: ortho and para.

40 a) (20 points) In para-hydrogen, the two electrons of H_2 form a singlet (antisymmetric) state. Therefore, the orbital angular momentum can have only even values. That is, the Hamiltonian has the form:

$$H = \frac{\hbar^2}{2I} l(l+1)$$

for $l = 0, 2, 4, \dots$

In contrast, in ortho-hydrogen, the electrons are in a triply degenerate symmetric state. Therefore, the orbital angular momentum can have only odd values. That is, the Hamiltonian has the form:

$$H = \frac{\hbar^2}{2I} l(l+1)$$

for $l = 1, 3, 5, \dots$

Compute the rotational partition function for both para-hydrogen and ortho-hydrogen, and evaluate both of their low and high temperature limits.

For para hydrogen,

$$Z_{\text{rot}} = \sum_{l=0}^{\infty} (2l+1) e^{-\frac{\beta \hbar^2}{2I} l(l+1)}$$

for even values, $l = 2n$

$$Z_{\text{rot}} = \sum_{n=0}^{\infty} [2(2n)+1] e^{-\frac{\beta \hbar^2}{2I} [2n(2n+1)]}$$

$$Z_{\text{rot}} = \sum_{n=0}^{\infty} (4n+1) e^{-\frac{\beta \hbar^2}{I} [n(2n+1)]}$$

At low temperature,

$T \rightarrow 0, \beta \rightarrow \infty, e^{-\infty} \rightarrow 0$

$$Z_{\text{rot}} = 1 + 5 e^{-\frac{3\beta \hbar^2}{I}} \checkmark$$

At high temperature

$4n+1 \rightarrow 4n, 2n+1 \rightarrow 2n$

$$Z_{\text{rot}} = \int_0^{\infty} 4n e^{-\frac{2\beta \hbar^2}{I} n^2} dn, \text{ let } b^2 = \frac{2\beta \hbar^2}{I}, b = \sqrt{\frac{2\beta \hbar^2}{I}}$$

$$Z_{\text{rot}} = \int_0^{\infty} 4(bn) e^{-b^2 n^2} d(bn) \frac{1}{b^2} = \frac{I}{2\beta \hbar^2} \int_0^{\infty} 4y e^{-y^2} dy$$

where $y = bn$

$$\frac{dy^2}{dy} = 2y, dy^2 = 2y dy, y dy = \frac{dy^2}{2} \checkmark$$

$$Z_{\text{rot}} = \frac{I}{2\beta \hbar^2} \int_0^{\infty} \frac{1}{2} e^{-y^2} dy^2 = \frac{I}{\beta \hbar^2} \int_0^{\infty} e^{-y^2} dy^2 = \frac{I}{\beta \hbar^2}$$

$$\text{since } \int_0^{\infty} e^{-y^2} dy^2 = 1$$

For ortho hydrogen

$$Z_{\text{rot}} = 3 \sum_{l=0}^{\infty} (2l+1) e^{-\frac{\beta \hbar^2}{2I} l(l+1)}$$

for odd values, $l = 2n+1$

$$Z_{\text{rot}} = 3 \sum_{n=0}^{\infty} [2(2n+1)+1] e^{-\frac{\beta \hbar^2}{2I} (2n+1)(2n+2)}$$

$$Z_{\text{rot}} = 3 \sum_{n=0}^{\infty} (4n+3) e^{-\frac{\beta \hbar^2}{I} (2n+1)(n+1)}$$

At low temperature

$T \rightarrow 0, \beta \rightarrow \infty, e^{-\infty} \rightarrow 0$

$$Z_{\text{rot}} = 9 e^{-\frac{\beta \hbar^2}{I}} \checkmark$$

At high temperature

$4n+3 \rightarrow 4n, 2n+1 \rightarrow 2n, n+1 \rightarrow n$

$$Z_{\text{rot}} = 3 \int_0^{\infty} 4n e^{-\frac{2\beta \hbar^2}{I} n^2} dn, \text{ let } b^2 = \frac{2\beta \hbar^2}{I}, b = \sqrt{\frac{2\beta \hbar^2}{I}}$$

$$Z_{\text{rot}} = 12 \int_0^{\infty} (bn) e^{-b^2 n^2} d(bn) \frac{1}{b^2} = \frac{12I}{2\beta \hbar^2} \int_0^{\infty} y e^{-y^2} dy$$

where $y = bn$

$$\frac{dy^2}{dy} = 2y, dy^2 = 2y dy, y dy = \frac{dy^2}{2}$$

$$Z_{\text{rot}} = \frac{12I}{2\beta \hbar^2} \frac{1}{2} \int_0^{\infty} e^{-y^2} dy^2 = \frac{3I}{\beta \hbar^2} \int_0^{\infty} e^{-y^2} dy^2 = \frac{3I}{\beta \hbar^2}$$

$$\text{Since } \int_0^{\infty} e^{-y^2} dy^2 = 1$$

- b (10 points) For an equilibrium gas of N hydrogen molecules composed of N_p para and $N_o = N - N_p$ ortho molecules, calculate the partition function of the complete system. (you may ignore vibrational degrees of freedom)

$$Z_{\text{tot}} = \sum_{N_o, N_p} \frac{1}{N_o! N_p!} Z_p^{N_p} Z_o^{N_o}$$

$$N_o = N - N_p$$

$$Z_{\text{tot}} = \sum_{N_p} \frac{1}{(N - N_p)! N_p!} Z_p^{N_p} Z_o^{N - N_p} = \frac{1}{N!} (Z_o + Z_p)^N \checkmark$$

$$\ln Z_{\text{tot}} = \ln \left[\frac{(Z_o + Z_p)^N}{N!} \right] = \ln (Z_o + Z_p)^N - \ln N! = N \ln (Z_o + Z_p) - \ln N!$$

$$\ln N! = N \ln N - N$$

- c (10 points) Compute an expression for the rotational contribution to the internal energy $\langle E_{\text{rot}} \rangle$. Determine its behavior at the high and low temperature limits. Note that due to the large transition times between ortho- and para-hydrogen, in most circumstances the mixture is not actually in equilibrium.

$$\langle E_{\text{rot}} \rangle = - \frac{\partial}{\partial \beta} \ln Z_{\text{tot}} = - \frac{\partial}{\partial \beta} [N \ln (Z_o + Z_p) - \ln N!] = - \frac{\partial}{\partial \beta} [N \ln (Z_o + Z_p) - N \ln N + N]$$

$$\langle E_{\text{rot}} \rangle = - \frac{\partial}{\partial \beta} [N \ln (Z_o + Z_p)]$$

At low temperature,

$$\langle E_{\text{rot}} \rangle = - \frac{\partial}{\partial \beta} [N \ln (1 + 9e^{-\frac{\beta \hbar^2}{I}})] = - N \frac{\partial}{\partial \beta} \ln (1 + 9e^{-\frac{\beta \hbar^2}{I}})$$

$$\langle E_{\text{rot}} \rangle = - N \frac{\partial}{\partial \beta} (1 + 9e^{-\frac{\beta \hbar^2}{I}}) \text{ as } \ln (1 + 9e^{-\frac{\beta \hbar^2}{I}}) \approx 1 + 9e^{-\frac{\beta \hbar^2}{I}}$$

$$\langle E_{\text{rot}} \rangle = - N \left(\frac{\hbar^2}{I} \right) 9e^{-\frac{\beta \hbar^2}{I}} \approx \frac{N \hbar^2}{I} 9e^{-\frac{\beta \hbar^2}{I}} \checkmark$$

At high temperature,

$$\langle E_{\text{rot}} \rangle = - \frac{\partial}{\partial \beta} [N \ln (\frac{I}{\beta \hbar^2} + \frac{3I}{\beta \hbar^2})] = - N \frac{\partial}{\partial \beta} \ln (\frac{4I}{\beta \hbar^2})$$

$$\langle E_{\text{rot}} \rangle = - N \frac{\partial}{\partial \beta} [\ln (4I) - \ln \beta - \ln \hbar^2]$$

$$\langle E_{\text{rot}} \rangle = - N \left(- \frac{1}{\beta} \right) = \frac{N}{\beta} = k_B T N = N k_B T \checkmark$$

Have a good break!